

UML Object-Oriented Analysis & Design

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Assessment Item: Question 9 (Sub-parts q and r)

Topic: Unified Modeling Language (UML) System Architecture Diagrams

9. q) Online Shopping Cart System

System Description: Develop a system using UML for implementing an Online Shopping Cart System. The customer browses the product catalogue and adds items to the cart. The system verifies product availability through the inventory module. The customer proceeds to checkout, makes payment through the payment gateway, and receives an order confirmation. The inventory is updated after each successful order.

1. USE CASE SPECIFICATION

This diagram captures the functional requirements of the system and the interactions between the system actors and the specific use cases.

Actor	Associated Use Cases	Description / Relationships
Customer (Primary)	• Browse Product Catalogue • Add Items to Cart • Proceed to Checkout • Make Payment	Initiates the shopping lifecycle, manages their selection, and concludes with checkout verification.
Inventory Module (Secondary)	• Verify Product Availability • Update Inventory	Triggered systematically via <<include>> relationships during cart addition and confirmation.
Payment Gateway (Secondary)	• Make Payment	External handler responsible for securely authorizing financial transactions.

2. CLASS DIAGRAM SPECIFICATIONS

The core structural backbone representing the attributes, methods, and static relationships within the Online Shopping Cart system.

Customer	Cart
+ int customerId + String name + String email	+ int cartId + List<Product> itemsList + double totalPrice
+ browseProducts() + checkout()	+ addItem(Product p) + removeItem(Product p) + calculateTotal()

Product	Order
+ int productId + String name + double price + int stockQuantity	+ int orderId + String status + double finalAmount
+ updateStock(int qty)	+ createOrder() + sendConfirmation()

3. BEHAVIORAL SEQUENCE TIMELINE

The exact sequential flow of messages exchanged between objects over time during a successful ordering operation:

- 1 Customer → Catalogue / Cart:** Requests to browse and appends selected items to the active Cart instance.
- 2 Cart → Inventory Module:** Sends a synchronous check to verify if requested quantities are physically in stock.
- 3 Inventory Module → Cart:** Returns acknowledgment message confirming stock availability status.
- 4 Customer → Checkout / Order System:** Initiates checkout step, compiling current items list into an implicit Order state.
- 5 Order System → Payment Gateway:** Calls external API containing transaction data to authorize payment.
- 6 Payment Gateway → Order System:** Returns an authorized receipt token signifying successful transaction processing.
- 7 Order System → Inventory Module:** Issues an update instruction to immediately decrement the product stock level.
- 8 Order System → Customer:** Renders successful order confirmation view and dispatches validation summary.

9. r) ATM Banking System

System Description: Develop a system using UML for implementing an ATM Banking System. The customer inserts the card and enters the PIN which is verified by the central banking server. Upon successful authentication the customer can check balance, withdraw cash, or transfer funds. The system records each transaction and updates the account database accordingly.

1. USE CASE SPECIFICATION

This details the interactive processes boundary accessible across an automated teller machine terminal system.

Actor	Associated Use Cases	Description / Relationships
Customer (Primary)	• Insert Card & Enter PIN • Check Balance • Withdraw Cash • Transfer Funds	The authenticated entity operating the physical hardware terminal unit.
Central Banking Server (Secondary)	• Verify PIN <<include>> • Update Account Database	Authoritative multi-tier backbone handling authentication lookups and state management.

2. CLASS DIAGRAM SPECIFICATIONS

The object relationships, multiplicity mapping, and modular components underpinning the physical-to-digital banking layout.

ATMMachine	Card
+ int atmId + String location + double cashAvailable	+ String cardNumber + int encryptedPin
+ insertCard() + verifyPIN(int pin) + dispenseCash(double amt)	+ getCardDetails()

Account	Transaction
+ String accountNumber + double balance	+ int transactionId + String type + double amount + Timestamp time
+ getBalance() + credit(double amt) + debit(double amt)	+ logTransaction()

3. BEHAVIORAL SEQUENCE TIMELINE

The sequence interaction dynamic tracking state changes across secure network nodes:

- 1 **Customer** → **ATM Machine**: Inserts debit card token and feeds input PIN payload keys.
- 2 **ATM Machine** → **Central Server**: Relays standard validation envelope request with credentials.
- 3 **Central Server** → **ATM Machine**: Verifies token records and signals a success authentication clear token.
- 4 **Customer** → **ATM Machine**: Declares action selection type (e.g., Check Balance, Withdraw, Transfer).
- 5 **ATM Machine** → **Central Server**: Transmits parameterized transaction request payload packet.
- 6 **Central Server** → **Account Database**: Commits transactional debit/credit modifications inside atomic block updates.
- 7 **Account Database** → **Central Server**: Dispatches success acknowledgment confirmation flag.
- 8 **Central Server** → **ATM Machine**: Commands physical mechanical drawer assembly to distribute funds or display receipt state info.
- 9 **ATM Machine** → **Customer**: Hands over local bills payout and returns physical card asset safely.